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Tripping on Aesthetascs:

A pilot study to characterize the impacts of thermal stress on *Hemigrapsus oregonensis*
In the presence of European green crab, *Carcinus maenas.*

ABSTRACT

In Pacific Northwest intertidal ecosystems, the increased presence of invasive European green crab (*Carcinus maenas*) is a cause of rising concern; *C. maenas* presence disrupts spatial and resource use dynamics, and often disturbs valuable habitat, producing additional predation and competition challenges for native shorecrabs. Olfaction, the primary sense used in foraging and predator avoidance, is one of many processes impacted by additional anthropogenic stressors, including ocean warming and acidification. By flicking their antennules, equipped with arrays of aesthetascs- delicate hair-like sensilla- crabs rely on sampling minute chemical cues in water to navigate the intertidal zone. However, few studies have considered potential isolated effects of temperature on the reception of chemical cues. Here, we present a framework to examine the effects of heightened temperatures on chemoreception in the native hairy shorecrab, *Hemigrapsus oregonensis. H. oregonensis* were reared at control and elevated temperatures, then exposed to food cues and the presence of live green crab, in order to evaluate the impact of thermal stress on olfactory function. This pilot study offers a template for the future study of crustacean physiology under stress, crucial for understanding invaded aquatic ecosystems as habitats shift under anthropogenic pressures.

INTRODUCTION

In recent years, conservationists and scientists have been battling a rapid marine invasion: the prolific invader, European green crab (*Carcinus maenas),* has been introduced to shorelines from California to British Columbia (Dunagan 2025). Aquatic invasions can affect native biota in a variety of ways, ranging from direct predation and food web disruptions to impacts on parasite load, reproductive success, and even mutualistic interactions across species (Jensen et al. 2002). Coupled with the effects of anthropogenic climate change and other human disturbances in the intertidal, these disruptions can be catastrophic. Green crab settlement, in particular, is known to have destructive effects on key habitat types such as eelgrass beds. In many ecosystems worldwide, their arrival introduces a dominant resource competitor for other crabs, as well as a formidable and unfamiliar predator for intertidal organisms. As the *C. maenas* invasion continues to spread throughout the state, a comprehensive understanding of its individual and population-level effects on resident native species is critical, in order to preserve the health of these diverse coastal communities.

In this pilot study, we investigate the impacts of thermal stress on the reception of food and predator-related olfactory cues in *Hemigrapsus oregonensis. H. oregonensis,* commonly known as hairy shore crab, is a small crab species found in abundance throughout the West coast of North America (Jensen et al. 2002). Although this intertidal crab can thrive under a variety of thermal and saline stressors, anthropogenic environmental changes, such as seawater acidification, have been shown to impair its chemosensory abilities (Khodikian et al. 2025). Decreased pH, in particular, is known to affect the structure of both cue-carrying molecules and the cue receptors themselves, disrupting the flow of information; however, the isolated effects of temperature on the chemosensory system have not been studied in this species (Ohnstad et al. 2025).

Crab behavior often hinges on complex signals about an individual's environment and its inhabitants: this includes chemical cues which signal the presence of food, mates, competing species, or predators (Fletcher & Hardege 2009). In crabs, olfaction is carried out by specialized chemosensory organs: flexible sensilla found on the antennules, which are moved, or "flicked" through the surrounding water (Waldrop 2013). The flicking motion passes water across a regularly spaced array of hair-like "aesthetascs" on each antennule; each aesthetasc has a thin cuticle which acts as a barrier, permeable to some odors and ions which carry environmental signals (Waldrop 2015). A single sensory hair contains hundreds of bipolar neurons, which contain sensory cilia, or hairs. These hairs are permeable, bipolar, and can respond to individual or complex stimuli. (Rittschof 1992). Therefore, each flick can be understood as a discrete intermittent sampling cycle.

As heightened antennule flicking corresponds to a higher stress response in *H. oregonensis*, flicking frequency was chosen to qualify crabs' awareness of a predator chemical cue in water. We predicted that with an elevated metabolic rate, *H. oregonensis* held in warmer temperatures would show elevated respirometry results throughout the testing period. If living under heightened temperatures impairs chemosensory ability, or if these temperatures impose strong energetic trade-offs for the species, we would expect to observe longer food-search times, and less flicking activity in the presence of green crab. We anticipated that crabs held in control temperatures would perform better under these trials, locating food more quickly and potentially responding more intensely to a predator.

METHODS

At the offset of the study, twelve *H. oregonensis* crabs of varying size and sex were removed from the holding tank, dried, weighed, and sized for carapace width . Each individual's carapace was then fitted with a small plastic tag, numbered accordingly, and a photo ID was taken. After 10 minutes of acclimation post-weighing, the crabs underwent their first Resazurin assay to establish baseline respirometry for each individual. Shorecrabs were placed in a well plate containing 8mL Resazurin solution, and T0 measurements were taken from each well, including one empty control well. Aliquot measurements were taken each half hour, for a total of 4 measurements per crab (T0-T4).

After handling, the *H. oregonensis* were separated into control and experimental groups: six control crabs were transferred to a cold-water tank held at 14°C, while the remaining six were introduced to a heightened temperature tank at 24°C. Once established, they were left in separate tanks, each with rocky substrate, for a week in order to adjust to the temperatures. Crabs were fasted for three days before both rounds of experimental trials.

During the first week of trials, each individual was removed from the holding tank and placed in an experimental tank with a small plastic "maze"; two low walls provided a visual barrier from the food parcel, a piece of thawed herring, which was placed on the opposite end of the tank after crabs were allowed a minute of acclimation. Once the food cue was introduced to the tank, the crab's response was timed from start to finish, when the individual located and took hold of the fish parcel. If crabs did not locate the food within five minutes, the trial was ended.

In order to examine *H. oregonensis'* response to the *C. maenas* chemical cue, a separate experimental tank was used, with a full plastic barrier constructed to separate the two sides visually and prevent crabs from interacting, but maintain water flow beneath the wall. After a 5-10 minute recovery and a minute spent sitting in the second experimental tank, one European green crab was introduced to the opposite chamber in the tank. *H. oregonensis*' responses were then video recorded for a minute as they experienced the olfactory cue from the green crab's presence, during which the individual's flicking behavior was observed and qualified. After all crabs had completed both chemical cue trials, a second Resazurin assay was conducted. The assay was completed using the process detailed above in order to establish a second measure of respirometry, after metabolic rates had adjusted to the different temperature conditions.

Crabs were held for another week in the seawater tanks, and the full experimental process was repeated the next week; *H. oregonensis* underwent trials in both testing tanks with food and predator-related chemical cues, and followed by a third Resazurin assay. Before this week's trials, we discovered a mortality in our testing group: one of our six high-temperature crabs had died, and this may have affected the fasting status of the testing group, which fed on the body. Finally, haemolymph was extracted from each individual, and an exploratory protein assay, not presented below, was conducted on the samples.

FINDINGS

Although crabs in the high-temperature group initially showed slightly higher respiration relative to control crabs, after just one experimental week, differences in respirometry had stabilized between groups (figure 1). The experimental crabs, therefore, showed signs of acclimation to the warmer tank, as demonstrated by respiration rates comparable to the control group.

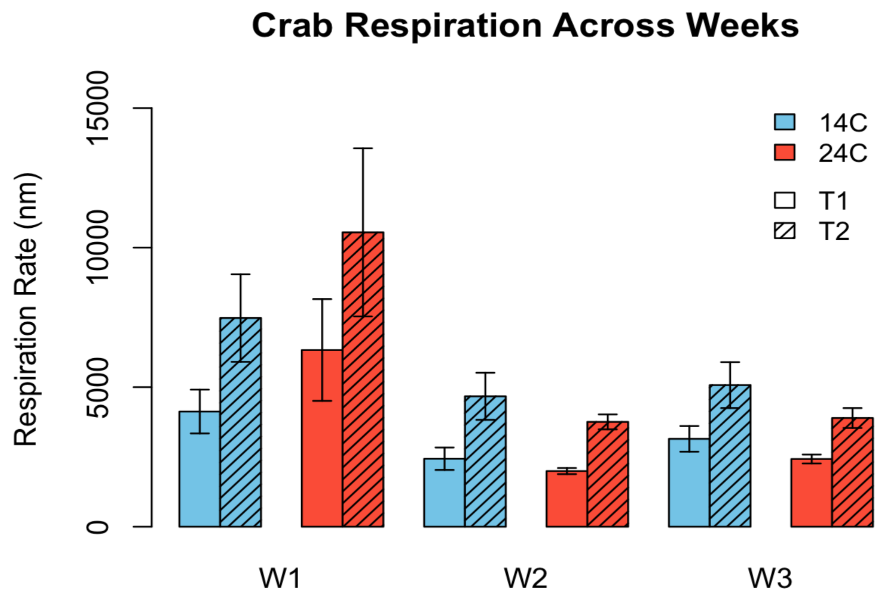


Figure 1: *Hemigrapsus oregonensis* recorded across our three testing periods. W1 is the first reading taken from our crabs before any trials, with W2 and W3 as weeks three and four. Crab 12 was only part of week 1 in the 24C group, since it died between week 1 and week 2.

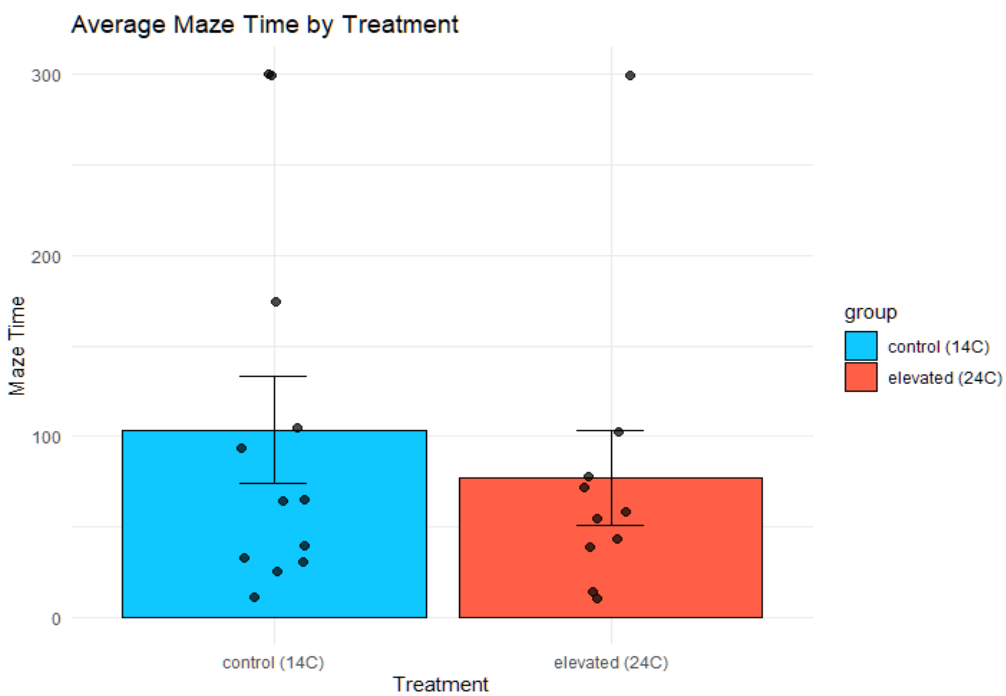
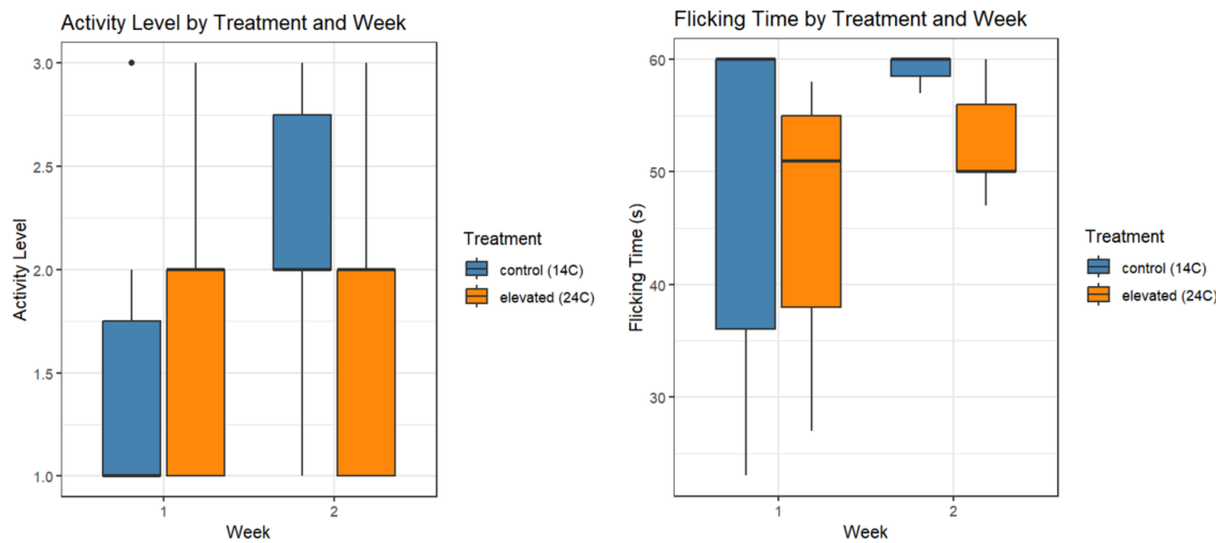


Figure 2: this graph shows the average time (in seconds) it took for crabs to find food in the maze with individual measurements as points overlaid on the bars. Individuals which spent 5 minutes in the maze were considered to have failed to find the food.

During the food cue trials, crabs held at elevated temperatures generally responded more quickly to the fish parcel. Across two weeks of testing, the 24°C group took a shorter time on average to locate food, although the difference was not statistically significant (figure 2).



Figures 3 and 4: these figures present the results for activity level and flicking time(left and right) by treatment group. Blue bars represent the control group, while orange bars show the 24°C group.

Similarly, there were no statistically significant differences in activity (flicking level) and flicking time between experimental groups, although control crabs showed slightly higher degrees of flicking than high-temperature crabs during the second week of trials (figure 3). Broadly, the two groups performed very similarly under both trials, with no evidence of heightened temperatures conferring any energetic cost to the crabs.

DISCUSSION

As shown by the plot of average respiration across weeks, both temperature groups had comparable respirometry rates throughout the testing period; although week one respiration was higher in the experimental set, the differences were not statistically significant. As all crabs were initially sourced from the same subject pool, the similarities in week one respiration rates are to be expected. However, the continuing similarity across groups is likely evidence of acclimation, as increased temperatures necessarily hasten metabolic rates (Clarke & Fraser 2004).

At the study offset, individual crabs were selected to randomize body size, and both mass and carapace width were randomly distributed across groups. Although this was the case, one individual in the testing group (crab #12) died before a second week of trials could be completed, decreasing the sample size to five. Since this crab was the largest of the high-temperature group, the average respiration rates may be deflated for this set of crabs. It also had an extremely high respiration rate (reflected in the chart error bar), which may be related to its subsequent death, and may have skewed the data for the 24C group. Its death may also have affected the maze time and flicking trials, as its tankmates consumed the body, and so were unfasted while the control group had gone without food. Other confounding factors, such as adjacent experimental setups in the lab, handling stress, and sex of individuals may also have affected the results presented above. Alternatively, it is possible that the high-temperature conditions were still comfortably within the species' window of thermal tolerance, and that higher temperatures, or the combined effects of decreased pH and other anthropogenic stressors, are necessary to impair chemoreception. One notable result was the diminished flicking time in the testing group when exposed to *C. maenas*; while the gap is statistically insignificant, this may suggest that higher temperatures affect the reception of less obvious environmental cues.

Future research is necessary to address the degree to which *H. oregonensis,* and other native species, actually receive green crab odor as a predator cue. Having been introduced relatively recently to these coasts, *C. maenas'* species-specific odor cue may not induce the same predation response in native shorecrabs. Future studies, with similar structure and a larger sampling pool, could offer valuable insight into antagonistic reactions between these two species under anthropogenic stressors. Interestingly, acidified conditions have been shown to increase attraction to microplastics as food in European green crab (Ohnstad et al. 2025); these results may be reproducible for *H. oregonensis*, and pH, known to affect chemoreception broadly by changing molecular structures, likely also influences predator perception and avoidance.

Human-made disturbances, including ocean warming, acidification, and the spread of invasive species now affect coastal regions more severely than ever, and ever more rapidly. With these environmental shifts in progress, physiological knowledge of both native and invasive species is critical to predict, and work to offset, these challenges; this pilot study presents a template for future research, guiding a deeper understanding of chemoreception in *H. oregonensis*, and an exploration of complex interspecific dynamics in the invaded intertidal.

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